```
> restart; #Redshift;
>
> eq1 := 1/c^2 diff(f(r, t), t, t) - diff(f(r, t), r, r) = 0;
```

$$eq1 := \frac{\frac{\partial^2}{\partial t^2} f(r,t)}{c^2} - \frac{\partial^2}{\partial r^2} f(r,t) = 0 \tag{1}$$

```
> iv1 := f(2· Rs, 0) = 0, D[1](f)(0, 0) = -Rs;
```

$$iv1 := f(2\,Rs, 0) = 0, \mathrm{D}_1(f)(0,0) = -Rs \tag{2}$$

```
> Sol := pdsolve([eq1, iv1]);
```

$$Sol := f(r,t) = \frac{\left(_c_2\, t^2 + 4\,Rs^2 - 2\,Rs\,r + 2\,c_3\, t\right) c^2 + _c_2 \left(-4\,Rs^2 + r^2\right)}{2\,c^2} \tag{3}$$

```
> pdsolve([eq1, iv1])/(4·Pi·r^2);
```

$$\frac{f(r,t)}{4\,\pi\,r^2} = \frac{\left(_c_2\, t^2 + 4\,Rs^2 - 2\,Rs\,r + 2\,c_3\, t\right) c^2 + _c_2 \left(-4\,Rs^2 + r^2\right)}{8\,\pi\,r^2\,c^2} \tag{4}$$

```
> f1 := subs(t = 0, %)
```

$$f1 := \frac{f(r,0)}{4\,\pi\,r^2} = \frac{\left(4\,Rs^2 - 2\,Rs\,r\right) c^2 + _c_2 \left(-4\,Rs^2 + r^2\right)}{8\,\pi\,r^2\,c^2} \tag{5}$$

```
> f1 := (4 Rs^2 - 2 Rs r) c^2 + _c_2 (-4 Rs^2 + r^2)/(8 π r^2 c^2);
```

$$f1 := \frac{\left(4\,Rs^2 - 2\,Rs\,r\right) c^2 + _c_2 \left(-4\,Rs^2 + r^2\right)}{8\,\pi\,r^2\,c^2} \tag{6}$$

```
> h := series(f1, r);
```

$$h := \frac{4\,Rs^2\,c^2 - 4\,_c_2\,Rs^2}{8\,\pi\,c^2}\, r^{-2} - \frac{1}{4}\,\frac{Rs}{\pi}\, r^{-1} + \frac{_c_2}{8\,\pi\,c^2} \tag{7}$$

```
> subs(r = 2·Rs, h);
```

$$\frac{4\,Rs^2\,c^2 - 4\,_c_2\,Rs^2}{8\,\pi\,c^2}\, 2\,Rs^{-2} - \frac{1}{4}\,\frac{Rs}{\pi}\, 2\,Rs^{-1} + \frac{_c_2}{8\,\pi\,c^2} \tag{8}$$

```
> solve(int((4 Rs^2 c^2 - 4 _c_2 Rs^2)/(8 π c^2) ·2 ·Rs^-2 - 1/4 Rs/π ·2· Rs^-1 + _c_2/(8 π c^2), r = 0
```

```
..2· Rs ) = 2· Rs )
```

$$\left\{Rs = Rs,\ c = c,\ _c_2 = -\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2\right\} \tag{9}$$

```
> subs( _c2 = -8/7 π c^2 + 4/7 c^2, (4 Rs^2 c^2 - 4 _c2 Rs^2)/(8 π c^2) r^-2 - 1/4 Rs/π r^-1
  + _c2/(8 π c^2) );
```

$$\frac{4\,Rs^2\,c^2 - 4\left(-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2\right)Rs^2}{8\,\pi\,c^2\,r^2} - \frac{Rs}{4\,\pi\,r} + \frac{-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2}{8\,\pi\,c^2} \tag{10}$$

```
> series( (4 Rs^2 c^2 - 4 (-8/7 π c^2 + 4/7 c^2) Rs^2)/(8 π c^2 r^2) - Rs/(4 π r) + (-8/7 π c^2 + 4/7 c^2)/(8 π c^2), r );
```

$$\frac{4\,Rs^2\,c^2 - 4\left(-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2\right)Rs^2}{8\,\pi\,c^2}\,r^{-2} - \frac{1}{4}\,\frac{Rs}{\pi}\,r^{-1} + \frac{-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2}{8\,\pi\,c^2} \tag{11}$$

```
> simplify( (4 Rs^2 c^2 - 4 (-8/7 π c^2 + 4/7 c^2) Rs^2)/(8 π c^2) r^-2 - 1/4 Rs/π r^-1
  + (-8/7 π c^2 + 4/7 c^2)/(8 π c^2) );
```

$$\frac{\left(-4\,\pi + 2\right)r^2 - 7\,Rs\,r + 16\,Rs^2\left(\pi + \frac{3}{8}\right)}{28\,r^2\,\pi} \tag{12}$$

```
> series( ((-4 π + 2) r^2 - 7 Rs r + 16 Rs^2 (π + 3/8))/(28 r^2 π), r )
```

$$\frac{4\,Rs^2\left(\pi + \frac{3}{8}\right)}{7\,\pi}\,r^{-2} - \frac{1}{4}\,\frac{Rs}{\pi}\,r^{-1} + \frac{-4\,\pi + 2}{28\,\pi} \tag{13}$$

```
> Rs := 8.209331978;
```

$$Rs := 8.209331978 \tag{14}$$

```
> evalf(int(4*(Pi+3/8)*Rs^2/(7*Pi)*r^(-2)-1/4*Rs/Pi*r^(-1)+(-4*Pi+2)/(28*Pi), r=0.431
..13.79));
#Local Universe comoving radial distance is 93 billion
lightyears;
```

93.02203568 **(15)**

```
> evalf(int(4*(Pi+3/8)*Rs^2/(7*Pi)*r^(-2)-1/4*Rs/Pi*r^(-1)+(-4*Pi+2)/(28*Pi), r=0.29..13.79));
#Local Universe with the JST 's galaxy discocoery JADES-GS-
z14-0 existed only 290 million years after the big bang not a
valid comoving distance;
```

141.3750603 **(16)**

```
> evalf(int(4*(Pi+3/8)*Rs^2/(7*Pi)*r^(-2)-1/4*Rs/Pi*r^(-1)+(-4*Pi+2)/(28*Pi), r=0.431
..13.79)); # The ago of the LU 13.79 billion years;
```

93.02203568 **(17)**

```
> #Thus comoving radial distance is 116 byllion lightyears of the
Local Universe;
```

```
> evalf(int(4*(Pi+3/8)*Rs^2/(7*Pi)*r^(-2)-1/4*Rs/Pi*r^(-1)+(-4*Pi+2)/(28*Pi), r=0.431
..13.4));#The Milky Way has 13.4 billion years;
```

92.99664477 **(18)**

```
> evalf(int(4*(Pi+3/8)*Rs^2/(7*Pi)*r^(-2)-1/4*Rs/Pi*r^(-1)+(-4*Pi+2)/(28*Pi), r=1.187
```

```
..13.4))); #comoving radial distance
```

$$30.04866710 \tag{19}$$

```
> (int((4*(pi+3/8)*Rs^2)/(7*pi)*r^(-2)-1/4*Rs/pi*r^(-1)+(-4*pi+2)/(28*pi),r));
```

$$-0.1201207224\,r - 0.6532778818\,\ln(r) - \frac{43.10719657}{r} \tag{20}$$

```
> evalf((4*Rs^2*(pi+3/8))/(7*pi)*r^(-2)-1/4*Rs/pi*r^(-1)+(-4*pi+2)/(28*pi));
```

$$\frac{43.10719658}{r^2} - \frac{0.6532778818}{r} - 0.1201207224 \tag{21}$$

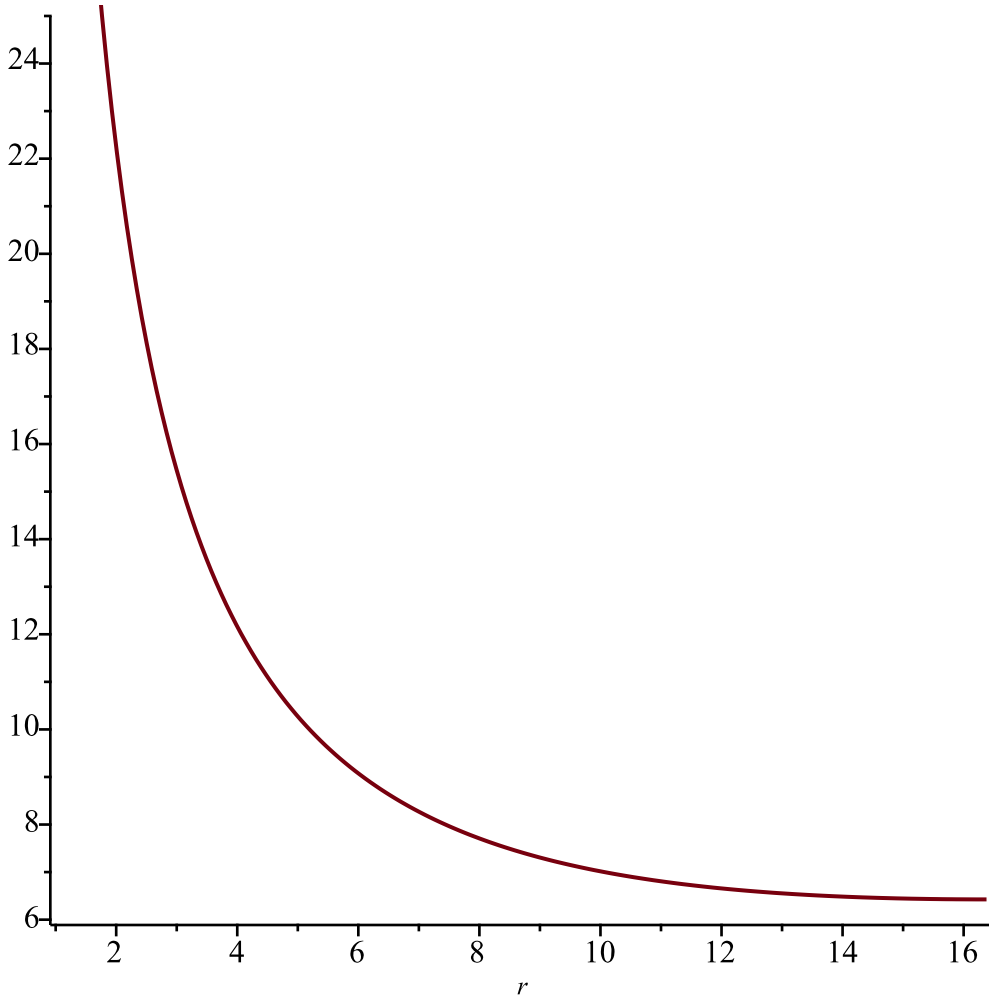
```
> L := evalf(-0.6532778818*ln(r)-0.1201207224*r-43.10719657/r);
```

$$L := -0.1201207224\,r - 0.6532778818\,\ln(r) - \frac{43.10719657}{r} \tag{22}$$

```
> evalf(int((4*(pi+3/8)*Rs^2)/(7*pi)*r^(-2)-1/4*Rs/pi*r^(-1)+(-4*pi+2)/(28*pi),r=0.929
..13.79));
```

$$39.96859485 \tag{23}$$

```
> plot(-L,r=0..2·8.20933197)
```




```
> restart;
```

```
> with(SolveTools[Inequality]) :
```

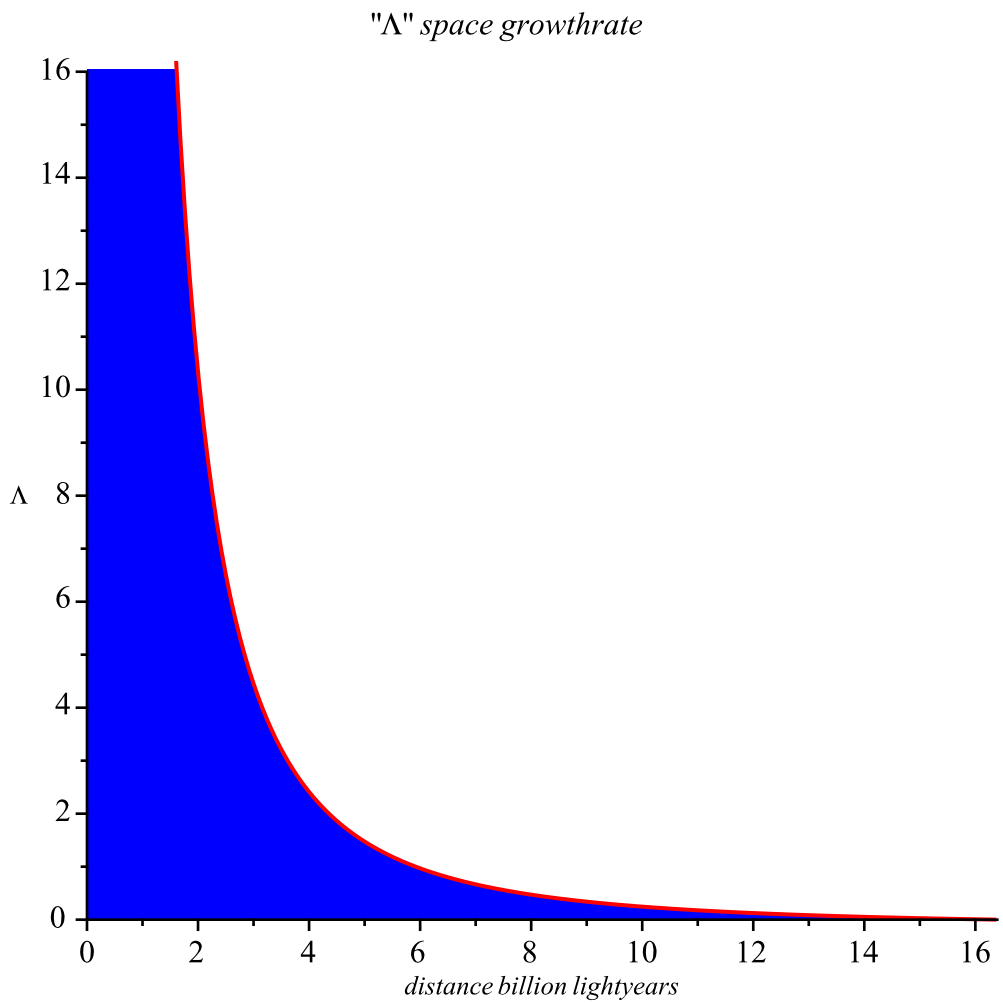
```
> c := 3;
```

$$c := 3 \tag{24}$$

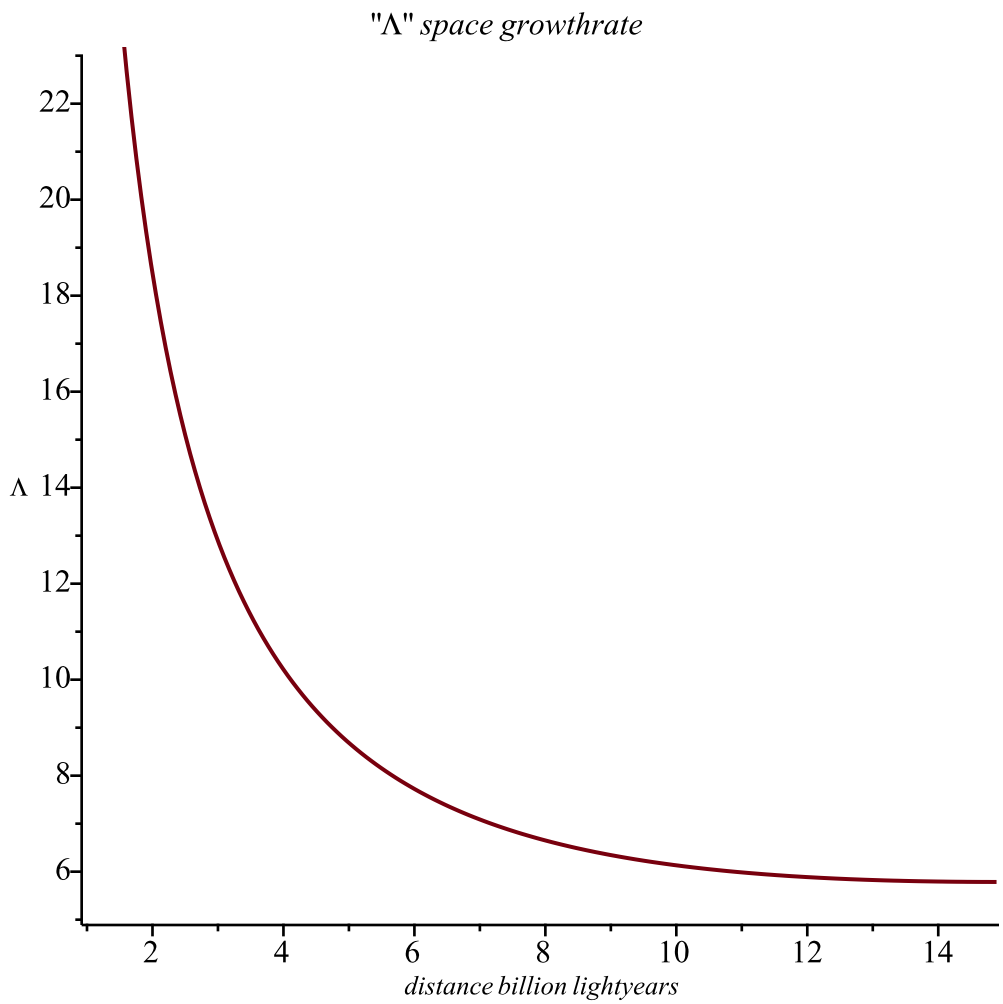
```
> Rs := 8.209331978;
```

$$Rs := 8.209331978 \tag{25}$$

```
> plot(((4 Rs^2 (π + 3/8))/(7 π)) r^(-2) - (1/4)(Rs/π) r^(-1) + (-4 π + 2)/(28 π)), r = 0..2·8.209331978,
    labels = [distance billion lightyears, Λ], filled = [color
    = "blue"], color = red, title = "Λ" space growthrate);
```




```
> plot( -( -0.5939475112 ln(r) - 0.1201207224 r - 35.63280450/r ), r = 0
  ..2·7.463764553, labels = [distance billion lightyears, Λ], title
  = "Λ" space growthrate );
```




```
> series( -r/7 + r/(14 π) - Rs ln(r)/(4 π) - 4 Rs^2/(7 r) - 3 Rs^2/(14 π r), r );
```

$$(-43.10719659)\, r^{-1} - 0.6532778818 \ln(r) - 0.1201207225\, r \tag{26}$$

```
> evalf( -r/7 + r/(14 π) - Rs ln(r)/(4 π) - 4 Rs^2/(7 r) - 3 Rs^2/(14 π r) );
```

$$-0.1201207225\, r - 0.6532778818 \ln(r) - \frac{43.10719659}{r} \tag{27}$$

```
> solve( -r/7 + r/(14 π) - Rs ln(r)/(4 π) - 4 Rs^2/(7 r) - 3 Rs^2/(14 π r) = c·t, t );
```

$$-\frac{1}{r}\left(3.031522726 \times 10^{-12}\left(7.183165480 \times 10^{10} \ln(r)\, r + 1.320796327 \times 10^{10} r^2 + 4.739883824 \times 10^{12}\right)\right) \tag{28}$$

```
> solve( [ -1/r (3.031522726 × 10^-12 (7.183165480 × 10^10 ln(r) r
```

```
    + 1.320796327 × 10^10 r^2 + 4.739883824 × 10^12)) = 0, 0 < r], [r]) ;
    #There is no solution that annulate space
```

$$[\,]$$ (29)

```
> restart;#neutrino Muon;
```

```
> de := diff(g(r), r$2) = r * g(r)/Rs;
```

$$de := \frac{d^2}{dr^2}\, g(r) = \frac{r\, g(r)}{Rs}$$ (30)

```
> ics := g( 2· Rs) = 0 , D(g)(0) = Rs;
```

$$ics := g(2\,Rs) = 0, \mathrm{D}(g)(0) = Rs$$ (31)

```
> sol := dsolve({de, ics});
```

$$sol := g(r) = \left(2\,Rs\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) + 3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\right)\right) - \left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) Rs\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\right)$$

```
> #Rs:=7.46;
```

```
> g(r) := (2 Rs π AiryBi(-2 (-1/Rs)^(1/3) Rs) AiryAi(-(-1/Rs)^(1/3) r)) /
    ((-1/Rs)^(1/3) Γ(2/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(
    -2 (-1/Rs)^(1/3) Rs))) - (2 AiryAi(-2 (-1/Rs)^(1/3) Rs) Rs π AiryBi(-(-1/Rs)^(1/3)
    -2 (-1/Rs)^(1/3) Rs)));
```

$$g := r \mapsto \left(2\cdot Rs\cdot\pi\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right) + 3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\right) - \left(2\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot Rs\cdot\pi\cdot\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right) \Big/$$ (33)

$$\left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\right)$$

```
> g(0);
```

$$\left(2\,Rs\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)\Big/\left(3\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)\right)-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)Rs\,\pi\,{}^{3}Rs\right)\right)$$

```
> # g(r) ·∂u(t)/∂t+u(t)· ∂g(r)/∂r·c=c;# thus for r=0 and t≠0 we
    have;
> dsolve(g(0)·diff(v(t), t) + v(t)·Rs·c − c = 0);
```

$$v(t)=\left(\frac{e^{-\frac{3\,c\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}-\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}}{Rs}+_C1\right)e^{-\frac{3\,c\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\,\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}} \tag{35}$$

```
> sol1 := v(t)·g(r);
```

$$sol1 := v(t)\left(\left(2\,Rs\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)\right)-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)Rs\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\left(-\right.\right.\right.$$

$$-\frac{1}{Rs}\Big)^{1/3} Rs\Big)\Big)\Big)\Big)$$

```
> H(r, t) := (1/Rs + e
  -(3 c Γ(2/3)^2 (-1/Rs)^(1/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(-2 (-1/Rs)^(1/3) Rs)) t)
   / (2 π (-AiryAi(-2 (-1/Rs)^(1/3) Rs) 3^(5/6) + AiryBi(-2 (-1/Rs)^(1/3) Rs) 3^(1/3)))

  C1) · (2 Rs π AiryBi(-2 (-1/Rs)^(1/3) Rs) AiryAi(-(-1/Rs)^(1/3) r)) /

  ((-1/Rs)^(1/3) Γ(2/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(

  -2 (-1/Rs)^(1/3) Rs))) - (2 AiryAi(-2 (-1/Rs)^(1/3) Rs) Rs π AiryBi(-(-1/Rs)^(1/3) r

  -2 (-1/Rs)^(1/3) Rs))));#_C1=C1;
```

$$H := (r, t) \mapsto \left(\frac{1}{Rs} + \mathrm{e}^{-\frac{3\cdot c\cdot\Gamma\left(\frac{2}{3}\right)^2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\cdot t}{2\cdot\pi\cdot\left(-\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{5/6}+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/3}\right)}}\cdot C1\right)$$

$$\cdot\left(\left(2\cdot Rs\cdot\pi\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\right.\right. \tag{37}$$

$$\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\right)-\left(2\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot Rs\cdot\pi\cdot\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\right)\right)$$

>